

Overtraining and Injury: Predicting Athlete Injury Risk from Performance and Training Data

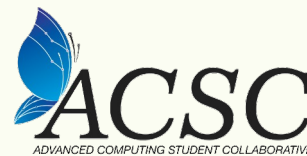
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Overview:

Overtraining is a major cause of preventable injuries

Focus questions:

- **How common** are overtraining-related problems?
- Do **training variables** predict injury risk?



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Methods:

- Four datasets analyzed:
 - 1 high-accuracy injury dataset, 3 activity datasets (walking, running, lifting) for injury prevalence
- Tools: Python, JupyterLab, TACC supercomputer
- Age groups created using **median split** for comparison



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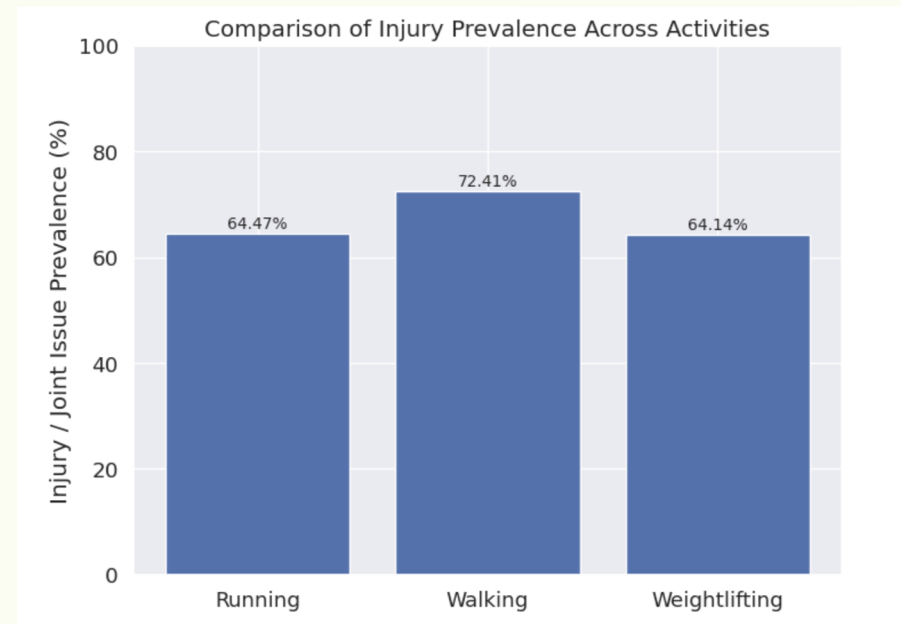
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Injury Prevalence Across Activities:

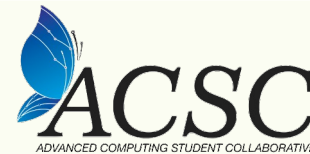
Key Points:

- Repetitive movement patterns cause overuse injuries
- Overtraining-related problems are widespread
- Even low-impact activities show high discomfort rates



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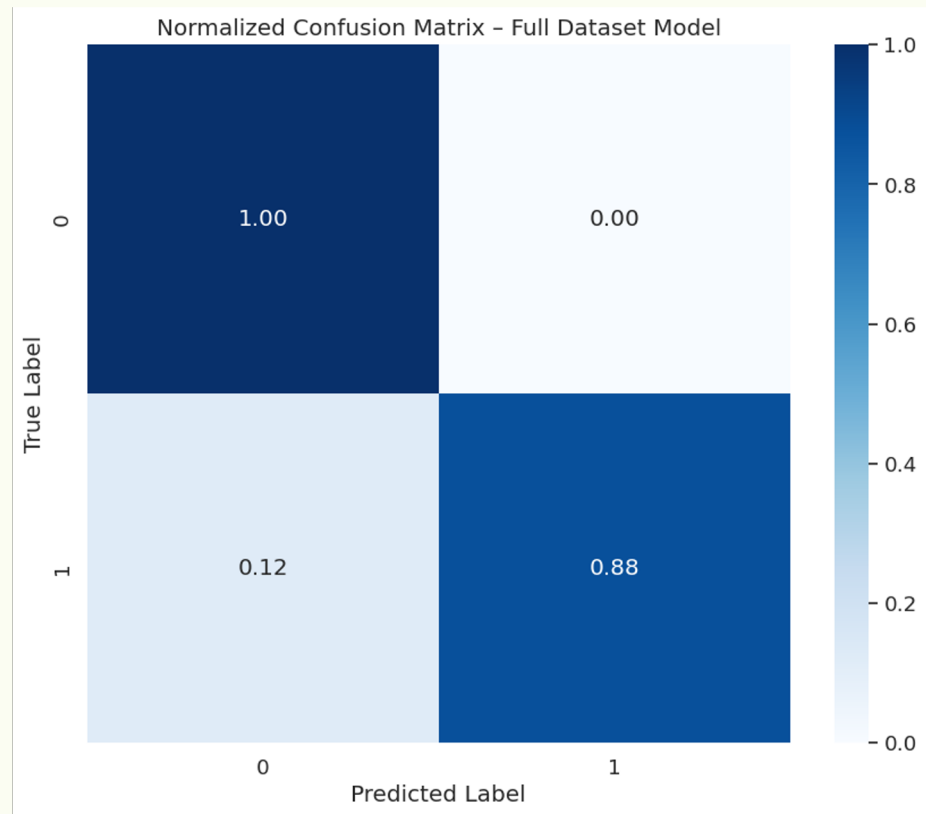
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Prediction Model Performance:

Key Points:

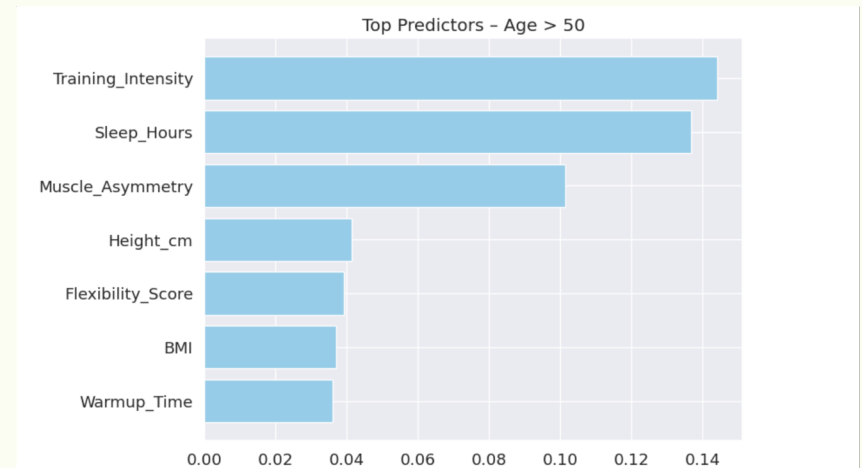
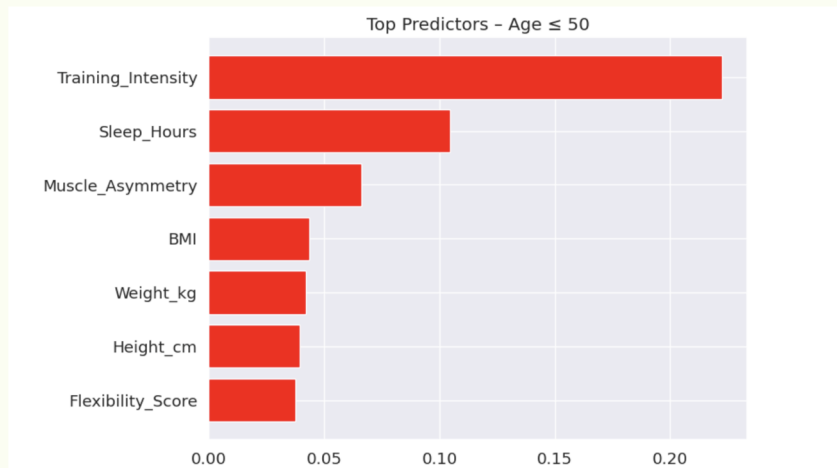
- Random Forest accuracy $\approx$ **97%**
- Strong diagonal = correct predictions
- Training variables contain **predictive patterns**



Feature Importance by Age Group:

Key Points:

- Younger athletes are influenced by training intensity & asymmetry
- Older athletes are influenced by recovery & load management
- Age changes how injuries develop



Conclusion:

- Injury prevalence is **high** across all activities
- Training variables are **strong predictors** of injury
- Machine learning can **identify risk before injury occurs**
- Age affects which variables matter most
- Potential for **early-warning systems** to reduce preventable injuries



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